INTEGRAL BIRATIONAL MORPHISMS OF DECENT SPACES IDENTIFY THE NORMALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5439-5446, immediately following the statement at 5431-5437.

Editorial note. The final identification (Y^nu inherits the normalization role from X^nu through the isomorphism) is compressed in the author text; the reading in proof_roadmap is the reviewer's checklist.

Editorial note. One cross-chapter pointer (Morphisms of Spaces) and two same-chapter line-indexed stubs are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Morphism induced on normalizations by a birational morphism. Lemma lemma-birational-induced-morphism-normalizations of this chapter, cited at proof lines 5439-5441. Supplies the map X^nu -> Y^nu over f whose integrality is to be shown.
- Integral morphisms are finite under decency hypotheses. Morphisms of Spaces, Lemma spaces-morphisms-lemma-finite-permanence, cited at proof lines 5442-5443. X^nu -> Y^nu integral (normalizations are integral; composition with f integral).
- Finite birational morphisms towards a normal scheme are isomorphisms. Lemma lemma-finite-birational-over-normal of this chapter, cited at proof lines 5444-5446. Y^nu is normal, so the finite birational X^nu -> Y^nu is an isomorphism.

Source reference. Original lemma 0BBG: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 5350–5350. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0B4E: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 5402–5402. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 081Z: focused-seeds/geometry/S9-decent-normalization-normal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an integral birational morphism of decent algebraic spaces over S which have finitely many irreducible components. Then there exists a factorization $Y^\nu \to X \to Y$ and $Y^\nu \to X$ is the normalization of X.*

Proof. Consider the map $X^\nu \to Y^\nu$ of Lemma 0BBG. This map is integral by Morphisms of Spaces, Lemma 081Z. Hence it is an isomorphism by Lemma 0B4E. □